STM 32 NUCLEO

PLATFORM

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1: Bare-Metal GPIO & Clock Enable via Pointer Structs

Problem:

On an STM32 Nucleo board, the user LED (LD2) is typically connected to PA5 (Port A, Pin 5). Write a bare-metal C program using structure pointers to.

1. Enable the clock for GPIOA in the RCC (Reset and Clock Control) register (RCC_AHB1ENR at offset 0x30 from base 0x40023800).
2. Configure PA5 as a General Purpose Output mode in GPIOx_MODER.
3. Toggle the LED every ~500ms using a simple software delay loop, accessing GPIOx_ODR directly.

2: EXTI Push-Button Interrupt Handler

Problem:

The Nucleo blue user button (B1) is connected to PC13. Configure PC13 as an external interrupt (EXTI13) triggered on the falling edge (button press connects pin to GND). Write the Interrupt Service Routine (ISR) using function pointers to toggle LD2 (PA5) whenever the button is pressed.

3: UART Ring Buffer via Pointer Arithmetic

Problem:

Nucleo ST-LINK virtual COM port connects to USART2 (PA2 = TX, PA3 = RX). Implement a circular RX ring buffer and an interrupt handler that stores incoming serial bytes using pointer indexing without dropping data during rapid typing.

4: Hardware Timer PWM for LED Breathing

Problem:

Configure TIM2 Channel 1 (mapped to PA5 on many Nucleo boards) to generate a Hardware PWM signal. Use pointer iteration inside a main loop to dynamically adjust the capture/compare register (CCR1), creating a smooth LED breathing effect.

5: ADC Single-Channel Polling with Register Structs

Problem:

Configure ADC1 to read an analog voltage on A0 (PA0). Write a polling function using pointer dereferencing that waits for conversion completion (EOC flag) and returns the raw 12-bit ADC result (0–4095).

6: SysTick High-Precision Microsecond Delay

Problem:

The ARM Cortex-M SysTick timer is a system timer built into the core (0xE000E010). Write initialization and delay functions using SysTick pointer registers (CTRL, LOAD, VAL) to achieve accurate microsecond (delay_us) delays.

7: UART Direct String Transmission

Problem:

Write a blocking function uart_send_string(const char *str) that transmits a null-terminated string over USART2 using pointer traversal and checking the Transmit Data Register Empty (TXE) bit in USART_SR.

8: Independent Watchdog (IWDG) Timer Refresh

Problem:

Embedded systems require watchdog timers to recover from lockups. Configure the STM32 Independent Watchdog using its register base (0x40003000) with a ~1-second timeout, and write a refresh function using key register (IWDG_KR) pointer writes.

9: Flash Memory Unlock & Sector Erase Sequence

Problem:

STM32 flash memory requires a strict key unlock sequence before modification. Write a function using pointers to unlock the Flash Controller (FLASH_KEYR at 0x40023C04) and erase Sector 7 (0x08060000) in flash memory

10: Memory-Mapped SPI Configuration & Transmission

Problem:

Configure SPI1 in Master mode (0x40013000). Write a pointer-based function spi_transfer_byte(uint8_t data) that sends a byte over MOSI while reading the incoming MISO byte from the data register (DR).